INSTALLATION INSTRUCTIONS

Power Glove Vision Installation Guide

Build, pair, and play with camera-based hand controls on Arduino UNO Q and RetroPie.

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Build, pair, and play with camera-based hand controls on an Arduino UNO Q and RetroPie.

PowerGlove Vision turns a bare hand - or a plain tracking glove - into a real Linux gamepad. The UNO Q handles the camera and gesture recognition. RetroPie receives authenticated controller packets and presents them to RetroArch as a virtual joystick.

> **THE SHORT VERSION** Camera into UNO Q. UNO Q and RetroPie on the same trusted network. Pair once. Map the virtual controller once. Then practise at `/learn` and play.

Recommended: one setup command per machine

First download the project to RetroPie, and import/run the application once in Arduino App Lab on the UNO Q. Connect both machines to your network. Then run the appropriate command **on that machine**, from the project/app directory. Replace the example hostnames with your own.

On RetroPie:

```sh
sudo python3 scripts/setup-machine.py retropie --peer UNO-Q-NAME.local
```

This installs the system Python receiver dependencies, Avahi and `libnss-mdns`, enables Avahi at boot, loads uinput, installs the delayed receiver timer, adds launch hooks and the controller profile, and creates only missing configuration. Existing pairing tokens, game registries, launcher settings and controller profiles are preserved. If changing an existing UNO Q destination, edit `/etc/powerglove/launcher.json` deliberately; `--peer` sets the initial destination only. Existing hook commands remain in place. Changed managed files are backed up under `/var/backups/powerglove-vision/`.

On the UNO Q, from `/home/arduino/ArduinoApps/powerglove-vision`:

```sh
sudo python3 scripts/setup-machine.py uno-q
```

This completes an App Lab import: Avahi and `libnss-mdns` for host name resolution, the persistent app-owned resolver service for container lookups, secure web port, shutdown helper, and default startup app. It restarts the application. This first version requires the standard app directory above. It does not change private device settings, pair without approval, or enable gameplay output. Select your RetroPie destination and pair through the Connection page afterward. New installations leave this destination blank and block controller output until it is configured. Learn mode, the website, and matrix animations remain available without pairing. Updates preserve an existing saved destination.

Both commands end with a report:

Result	Meaning
PASS	This installation check succeeded
FAIL	A technical problem needs correction
ACTION	A user step remains, such as pairing or verifying controls in a game

Exit codes are `0` for all checks passed, `1` for failure, and `2` for remaining user action. The report intentionally asks for gameplay verification; a token file or running process alone does not prove end-to-end control. No password or pairing token is printed. A missing token leaves the receiver waiting for pairing.

Rerun checks without installing, restarting, or modifying settings:

```sh
sudo python3 scripts/setup-machine.py retropie --check
```

```sh
python3 scripts/setup-machine.py uno-q --check
```

The detailed stages below remain available for manual installation and diagnosis. The installers do not install ROMs or BIOS files, choose a physical controller layout, or replace custom cabinet controller-merging software.

Choose your route

If you want to...	Go to...
Build the complete system for the first time	Stages 1-6, in order
Reconnect an existing build	Play Checklist
Pair the UNO Q and RetroPie	Stage 4
Configure automatic profiles	Stage 6
Understand every configuration file and field	`CONFIGURATION_REFERENCE.md`
Update over Wi-Fi	Workshop: updates and maintenance
Fix a camera, network, or controller problem	Troubleshooting
Understand Programs A-I	`bad-street-brawler-programs.md`

What you are building

```
                        trusted local network

UVC-compatible USB camera --USB--> UNO Q ===== authenticated UDP =====> RetroPie
                                    |                                     |
                                    | hand tracking                       | virtual gamepad
                                    | profiles + matrix                   | RetroArch / NES
                                    |
                                    +-- web workshop: Setup / Debug / Learn
```

The system does not require an original Power Glove or the Bad Street Brawler cartridge menu. A bare hand works. A plain white or black glove may improve contrast, but it contains no electronics.

Hardware bench

- Arduino UNO Q; the 4 GB model is recommended.
- Raspberry Pi running RetroPie and RetroArch.

- UVC-compatible USB camera. Power Glove Vision has been tested with a Razer Kiyo, but the Kiyo is not required.
- Externally powered USB-C hub or dock for the UNO Q and camera.
- Suitable 5 V/3 A UNO Q power supply.
- Computer with Arduino App Lab and a USB data cable.
- Shared trusted Wi-Fi or Ethernet network.
- Internet access during the UNO Q's first application start.

Network bench

Service	Direction	Port	Purpose
Controller state	UNO Q -> RetroPie	UDP 55355	Virtual gamepad input
Profile control	RetroPie -> UNO Q	UDP 55356	Per-game profile changes
Web workshop	Browser -> UNO Q	TCP 8088	Learn, Debug, ordinary Setup
Secure Setup	Browser -> UNO Q	TCP 8443	Password and code pairing
One-time pairing helper	UNO Q -> RetroPie	TCP 55357	Temporary code exchange

> **NETWORK SAFETY** Keep these ports on a trusted home network. Do not expose them through router port forwarding. Controller packets are authenticated; secure pairing traffic is encrypted with TLS.

Stage 1 - Put the UNO Q on Wi-Fi

1. Connect the UNO Q to the computer with a USB data cable.
2. Open Arduino App Lab and complete the board's initial setup.
3. Give the board a short, memorable name.
4. Join the same trusted Wi-Fi network used by RetroPie.
5. Apply available UNO Q and App Lab updates.
6. Record both the `.local` name and current IP address.

Test the friendly name from your computer:

```sh
SH
ping UNO-Q-NAME.local
```

Use the IP address as a fallback if `.local` discovery is unavailable. A router DHCP reservation makes that fallback stable.

> **KEEP IT PRIVATE** Wi-Fi passwords and board passwords belong in App Lab, never in `device.json`, a shell command, Git, or a screenshot.

Stage 2 - Install PowerGlove Vision on the UNO Q

Import and start the app

Clone the repository and build the App Lab installation ZIP:

```
output/app-lab/PowerGlove-Vision-Uno-Q.zip
```

```
SH
git clone https://github.com/mathan416/PowerGlove-Vision.git
cd PowerGlove-Vision
scripts/build-app-lab-package.sh
```

The generated App Lab installation ZIP is not stored in Git. It deliberately excludes Google's Hand Landmarker model. The first time an active gesture profile needs vision, the UNO Q downloads the model directly from Google into the app's persistent private `data/models/` directory and verifies its pinned SHA-256 checksum before starting tracking. **Gestures off** leaves the model and camera unopened. Later launches reuse that verified copy. A published GitHub release may provide the same model-free App Lab installation ZIP.

The repository retains a custom MediaPipe 0.10.18 ARM64 wheel because its dependency metadata is tailored for the headless UNO Q runtime. Do not replace it with the similarly named stock PyPI wheel without retesting the complete camera startup path. Sources, exact checksums, modifications, and applicable licenses are recorded in THIRD_PARTY_COMPONENTS.md.

In Arduino App Lab:

1. Import the App Lab installation ZIP as an Arduino App.
2. Open **PowerGlove Vision** and select **Run**.
3. Allow several minutes when first activating gestures. The app downloads and verifies the Hand Landmarker model, prepares an isolated Python 3.12 vision environment, and downloads its ARM64 dependencies once.
4. When the app is healthy, enable **Run at startup**.

The package contains the Linux vision service and the matrix sketch. Arduino's protected early boot display remains intact; PowerGlove Vision takes over the matrix after its application starts.

Connect the camera

Connect the powered hub to the UNO Q, then connect your UVC-compatible USB camera to the hub. The app automatically ignores the UNO Q's internal codec nodes and waits for a real USB camera. It is safe to connect the camera before or after the app starts.

> **POWER MATTERS** Webcam dropouts that look like software problems are often caused by an underpowered passive adapter. Use a powered hub and a solid cable.

Read the blue matrix

Display	Meaning
Animated hand	Application or model is loading
Animated glove with an energy sweep	Gestures are paused; Linux, the website, and profile selection remain available
`PG`	Ready; no specific profile selected yet
`A-I`	One of the cartridge-free Programs A-I is active
`BS`	Bad Street Brawler profile
`GB`	Super Glove Ball profile
Pulsing code	A calibrated hand is being tracked
Blinking X	Camera missing or vision worker needs attention

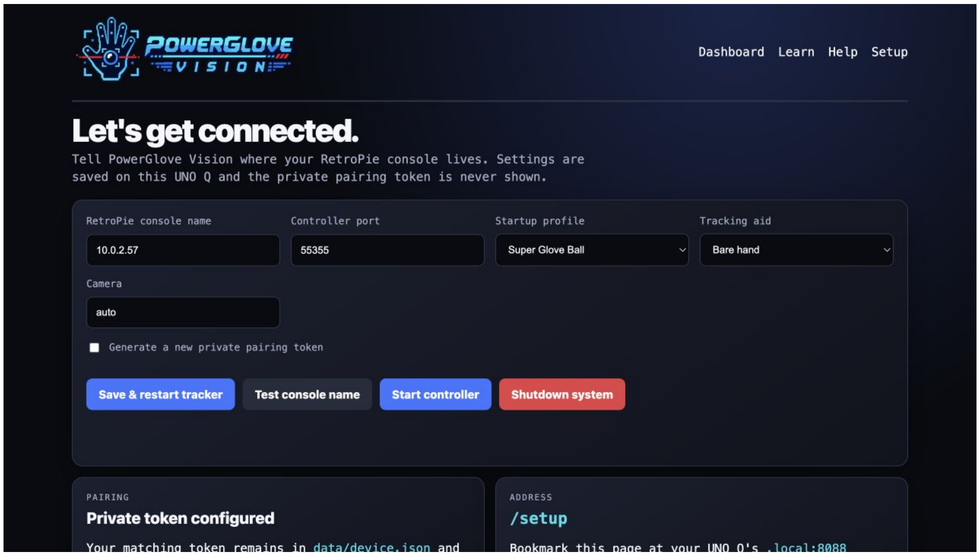
The web service remains available during a camera fault, so a blinking X is recoverable without reinstalling the app.

Confirm the web workshop

Open these pages from another device on the same network:

```
http://UNO-Q-NAME.local:8088/learn
http://UNO-Q-NAME.local:8088/debug
http://UNO-Q-NAME.local:8088/help
http://UNO-Q-NAME.local:8088/setup
https://UNO-Q-NAME.local:8443/setup
```

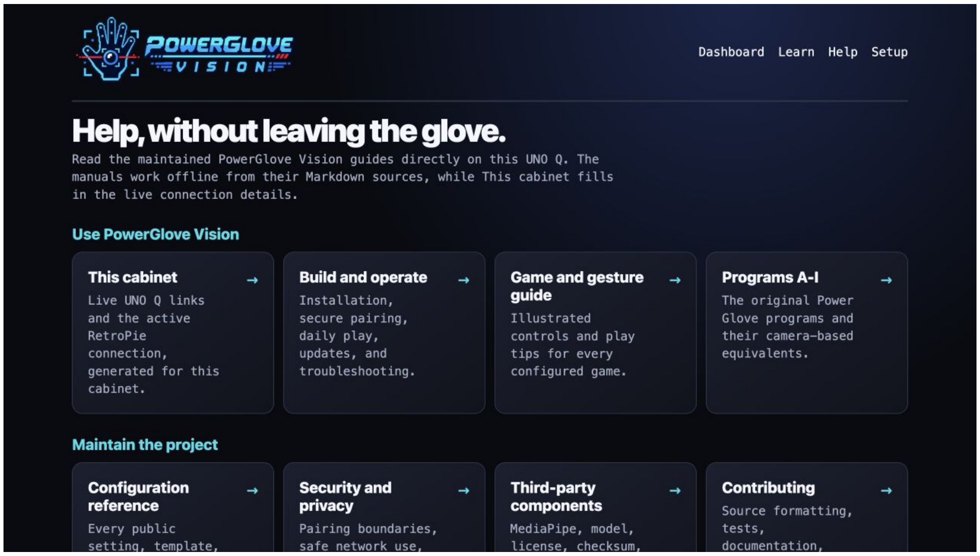
The browser may warn about the locally generated certificate on port 8443. That is expected; verify its fingerprint against the matrix during pairing.



PowerGlove Vision Setup page

The **Help** page renders the public Markdown manuals stored in `docs/` as an offline reading library. Choose a guide to get a styled reading view, guide navigation, a table of contents, illustrations, tables, and code samples. Use **View Markdown** when you need the original source, or **Open PDF** to read, save, or print the matching PDF edition. The Help landing page also provides the project overview PDF. These public PDFs work locally from the UNO Q without an Internet connection. The machine-specific `cheatsheet.md` and its quick reference PDF are deliberately excluded from the App Lab installation ZIP and do not appear in Help.

Choose **This cabinet** for a live quick reference without a static machine file. Its UNO Q links use the hostname or IP address that opened Help. Its RetroPie console, controller port, startup profile, tracking aid, camera, pairing readiness, and controller state come from the active public settings. The pairing token value and passwords are never returned.



PowerGlove Vision Help library

Stage 3 - Install the RetroPie receiver

Open a terminal on RetroPie, locally or through SSH.

Install system support

```sh
SH
sudo apt update
sudo apt install -y git python3 python3-venv python3-pip python3-dev \
  build-essential python3-evdev
sudo modprobe uinput
printf '%s\n' uinput | sudo tee /etc/modules-load.d/powerglove.conf
ls -l /dev/uinput
```

Install the project

```sh
SH
sudo git clone https://github.com/mathan416/PowerGlove-Vision.git /opt/powerglove-src
sudo python3 -m venv /opt/powerglove
sudo /opt/powerglove/bin/python -m pip install --upgrade pip
sudo /opt/powerglove/bin/python -m pip install -e '/opt/powerglove-src[receiver]'
sudo install -d -m 0755 /opt/powerglove/bin
sudo install -m 0755 /opt/powerglove-src/retropie/bin/* /opt/powerglove/bin/
```

On older RetroPie images, the system `python3-evdev` package avoids an obsolete PyPI build toolchain. The compatibility launchers above will use it.

Install configuration

```sh
SH
sudo install -d -m 0755 /etc/powerglove
sudo install -m 0644 /opt/powerglove-src/config/games.json /etc/powerglove/games.json
sudo install -m 0644 /opt/powerglove-src/config/launcher.example.json /etc/powerglove/launcher.json
sudo install -o root -g input -m 0640 /dev/null /etc/powerglove/token
sudo nano /etc/powerglove/launcher.json
```

Set `uno_q` to the board's hostname or reserved address:

```json
JSON
{
  "uno_q": "UNO-Q-NAME.local",
  "port": 55356,
  "token_file": "/etc/powerglove/token",
  "registry": "/etc/powerglove/games.json",
  "timeout": 0.4
}
```

Leave `/etc/powerglove/token` empty for now. Stage 4 fills it securely.

Stage 4 - Pair the two machines

Pairing gives the UNO Q and RetroPie the same private controller token. It is a one-time setup unless you regenerate the token or reinstall either machine.

Option A - One-time code (recommended)

On RetroPie:

```sh
SH
sudo /opt/powerglove/bin/powerglove-pair
```

Leave it running. It prints a single-use code valid for two minutes.

1. Open `https://UNO-Q-NAME.local:8443/setup`.
2. Enter the RetroPie hostname and its one-time code.
3. Select **Prepare one-time-code pairing**.
4. Watch the UNO Q matrix cycle through `ID`, fingerprint characters, `PN`, and a six-digit PIN.
5. Compare the matrix `ID` with the beginning of the browser certificate's SHA-256 fingerprint.
6. If they match, enter the matrix PIN and complete pairing.

The helper exits after success. It also expires after two minutes or five failed attempts.

Option B - RetroPie username and password

Use this when SSH password login is already enabled on RetroPie.

1. Open `https://UNO-Q-NAME.local:8443/setup`.
2. Enter the RetroPie hostname and username.

3. Select **Prepare password pairing**.
4. Compare the matrix `ID` with the browser certificate fingerprint.
5. Enter the fresh matrix PIN, confirm the fingerprint checkbox, and enter the RetroPie password.
6. Select **Pair RetroPie**.

The credentials are used for one encrypted SSH operation. The UNO Q does not store the password or place it in process arguments or logs. A private temporary token file is removed immediately after installation.

> **PAIRING PIN** Every prepared attempt receives a new PIN. If an attempt fails, prepare pairing again and use the new digits shown on the matrix.

Recovery - Manual token installation

Use this only when neither friendly pairing method is available. In App Lab, open the active app's private `data/device.json` and copy only its `token` value into `/etc/powerglove/token` on RetroPie, without quotation marks.

```
SH
sudo chmod 0640 /etc/powerglove/token
sudo systemctl restart powerglove-receiver.service
```

Never place the token in Git, documentation, screenshots, shell history, or an issue report. Selecting **Generate a new private pairing token** invalidates the old pairing; pair again immediately afterward.

Stage 5 - Start the virtual gamepad

Test packet delivery

```
SH
sudo /opt/powerglove/bin/powerglove-receiver \
  --listen 0.0.0.0 \
  --token-file /etc/powerglove/token \
  --dry-run
```

Show one hand to the camera, then select **Start controller** on Setup or Debug. Controller state should appear in the terminal. Press `Ctrl+C` when finished.

Install delayed receiver startup

Create `/etc/systemd/system/powerglove-receiver.service`:

```ini
INI
[Unit]
Description=PowerGlove Vision virtual gamepad receiver
Wants=network-online.target
After=network-online.target

[Service]
Type=simple
User=root
ExecStart=/opt/powerglove/bin/powerglove-receiver \
  --listen 0.0.0.0 \
  --token-file /etc/powerglove/token
Restart=on-failure
RestartSec=2

[Install]
WantedBy=multi-user.target
```

Install the supplied timer and verify it:

```sh
SH
sudo chmod 0644 /etc/systemd/system/powerglove-receiver.service
sudo install -m 0644 /opt/powerglove-src/retropie/powerglove-receiver.timer \
  /etc/systemd/system/powerglove-receiver.timer
sudo systemctl daemon-reload
sudo systemctl disable powerglove-receiver.service
sudo systemctl enable --now powerglove-receiver.timer
sudo systemctl start powerglove-receiver.service
sudo systemctl status powerglove-receiver.service
grep -A8 -B2 'PowerGlove Vision' /proc/bus/input/devices
```

The timer starts the receiver 45 seconds after boot. This keeps the virtual controller out of EmulationStation's initialization path; on the validated cabinet, starting the receiver too early caused frontend pauses and conflicts with other USB devices, including a BitPixel display. Do not enable the service directly in addition to the timer. The receiver creates its uinput device after its first authenticated packet.

Bind it in RetroArch

1. Open **Settings > Input > RetroPad Binds > Port 1 Controls**.
2. Select `PowerGlove Vision` as the device.
3. Bind D-pad directions, A, B, Start, and Select.
4. Save the controller profile or RetroArch configuration.

For automatic mapping:

```sh
SH
sudo install -m 0644 '/opt/powerglove-src/retropie/retroarch/PowerGlove Vision.cfg' \
  '/opt/retropie/configs/all/retroarch/autoconfig/PowerGlove Vision.cfg'
```

The extra `BTN_TR2` input preserves Bad Street Brawler's glove-zap event for a future native glove-aware emulator integration.

Stage 6 - Switch profiles with each game

RetroPie runcommand hooks can select the correct profile when a ROM starts and turn gestures off when it ends.

Add the hooks without replacing existing ones

```sh
sudo chmod 0755 /opt/powerglove-src/retropie/runcommand-onstart-powerglove.sh
sudo chmod 0755 /opt/powerglove-src/retropie/runcommand-onend-powerglove.sh
```

Append to `/opt/retropie/configs/all/runcommand-onstart.sh`:

```sh
/opt/powerglove-src/retropie/runcommand-onstart-powerglove.sh "$1" "$2" "$3" "$4"
```

Append to `/opt/retropie/configs/all/runcommand-onend.sh`:

```sh
/opt/powerglove-src/retropie/runcommand-onend-powerglove.sh
```

Make both cabinet hook files executable:

```sh
sudo chmod 0755 /opt/retropie/configs/all/runcommand-onstart.sh
sudo chmod 0755 /opt/retropie/configs/all/runcommand-onend.sh
```

> **CABINET ETIQUETTE** Append to existing hooks. Do not overwrite them; they may also manage lighting, bezels, controller order, or other cabinet hardware.

Review the game registry

`/etc/powerglove/games.json` matches exact ROM basenames case-insensitively. The shipped registry contains all eight games that Power Glove Vision recognizes and configures automatically out of the box:

Automatically recognized game	Profile
Bad Street Brawler	`bad_street_brawler`
Super Glove Ball	`super_glove_ball`
Joust	`program_b`
Gyruss	`program_c`
Defender II	`program_e`
Sesame Street 1-2-3	`program_f`
Gun Smoke	`program_g`
Knight Rider	`program_i`

Programs A, D, and H are also fully implemented. They are control profiles, not missing game entries, and are intentionally unassigned because none is tied to one specific title:

Available profile	Intended use	Default game assignment
`program_a`	Pinball controls with two finger flippers and wrist tilt	None
`program_d`	Reversed-direction challenge or accessibility experiments	None
`program_h`	General play and training with conventional movement	None

You can assign any appropriate NES or Famicom ROM to one of these profiles by adding its exact filename to `/etc/powerglove/games.json`. This is a configuration change and does not require new code. For example:

```json
JSON
{
  "games": {
    "YOUR EXACT PINBALL ROM FILENAME.nes": "program_a"
  }
}
```

Keep the existing entries when adding your own. An unknown game turns gesture control off instead of inheriting the previous game's profile.

Test a profile manually:

```sh
SH
sudo /opt/powerglove/bin/powerglove-profile \
  --uno-q UNO-Q-NAME.local \
  --token-file /etc/powerglove/token \
  --profile program_b
```

The command should acknowledge the change and the matrix should show B.

Play Checklist

1. Power the RetroPie and UNO Q; leave the camera connected to the powered hub.
2. Open `http://UNO-Q-NAME.local:8088/debug`.
3. Select the active profile on the Dashboard, then confirm the expected profile and a detected hand. The saved startup profile remains on Setup.
4. On first use, or after changing your camera or playing position, select **Calibrate** while holding a comfortable neutral pose. Otherwise reuse the saved calibration.
5. Select **Start controller** only when you are ready to play.
6. Launch the game and confirm its profile code on the matrix.

7 Select **Stop controller** before adjusting the camera or leaving the cabinet.

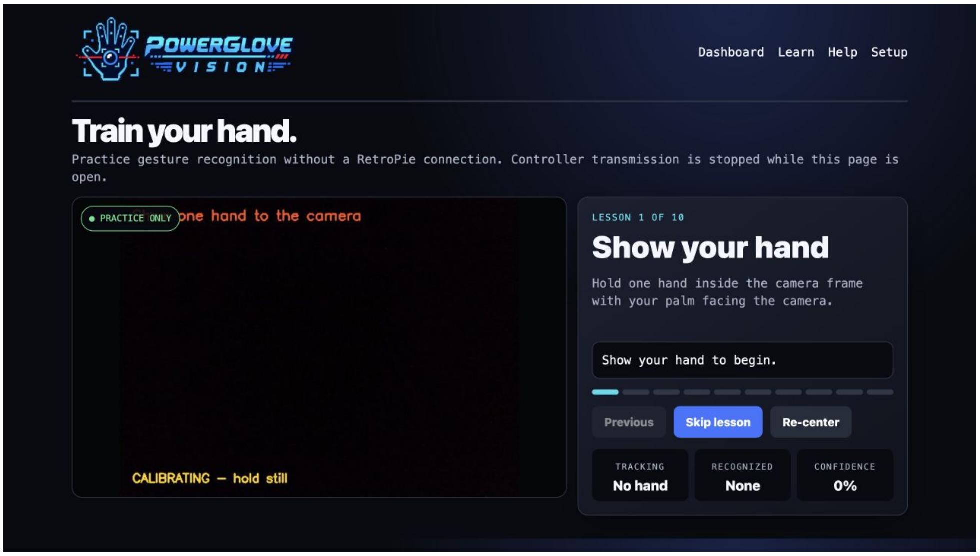
8 Read the shutdown limitation before disconnecting power. **Shutdown** requests a graceful halt, but the tested board restarts; an offline website is not proof that it is safe to unplug.

PowerGlove Vision deliberately boots with controller delivery stopped. Vision and the dashboard keep running so setup never generates surprise game inputs. **Shutdown** is different: it halts Linux on the UNO Q. The tested board automatically restarts; remaining halted is not guaranteed.

Learn before you launch

Dashboard and Learn show **Starting camera and gesture tracking** with elapsed seconds during initialization. The first activation can take longer while the camera and tracker load. Wait for vision to become active before centering; the centering button is disabled during startup. Gestures off keeps the camera off rather than briefly activating it at boot.

Open `http://UNO-Q-NAME.local:8088/learn`. Learn mode automatically stops controller transmission, starts the camera when necessary, and walks through ten exercises with live recognition feedback. RetroPie does not need to be online. Leaving Learn restores the selected profile and its camera state; loading or refreshing the Dashboard also reapplies the selected mode.

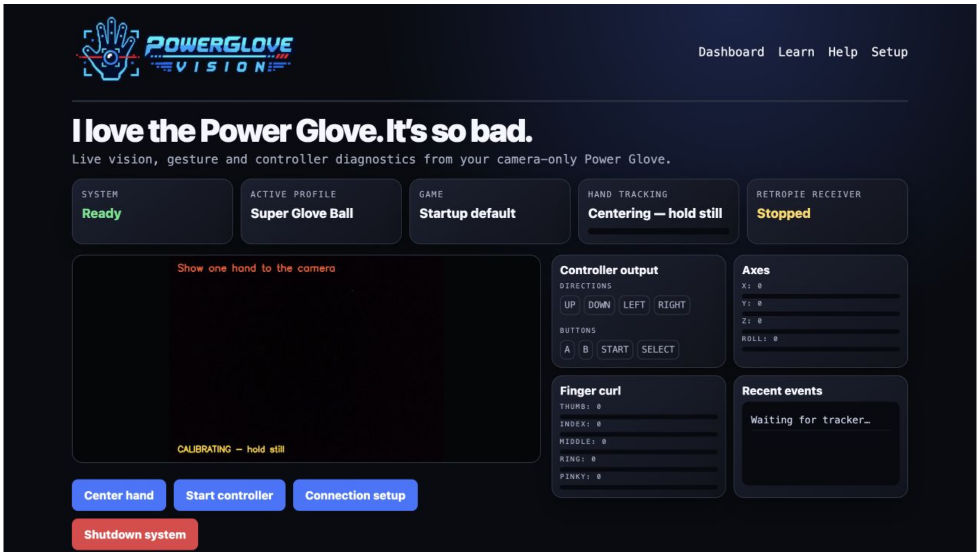


PowerGlove Vision Learn page

For a white glove, use a darker background. For a black glove, use a lighter background. Even lighting, a simple scene, and keeping the whole hand in frame matter more than glove color.

Read the live dashboard

Debug shows the camera overlay, active profile, tracking confidence, generated D-pad/buttons, analogue axes, finger curl, and recent gesture events. **Right** or **Left** in the camera overlay identifies the detected hand, not a movement direction. Its confidence score describes that handedness classification, not confidence in a movement command. Use the D-pad/buttons and axes to inspect actual generated controls. Its profile selector changes the current session immediately. Selecting **Gestures off** releases controls, closes the camera and shows a friendly idle panel; selecting another profile starts vision using the saved calibration. Recalibrate only when your camera or playing position changes, or neutral is incorrect.



PowerGlove Vision Debug dashboard

Workshop - Updates and maintenance

UNO Q updates over Wi-Fi

For repeatable developer deployments, install your computer's public SSH key for the UNO Q `arduino` account once over USB. Never copy the private key.

```sh
install -d -m 0700 /home/arduino/.ssh
chmod 0600 /home/arduino/.ssh/authorized_keys
```

Confirm key access from the development computer:

```sh
ssh -o BatchMode=yes arduino@UNO-Q-NAME.local hostname
```

Then deploy from the repository root:

```sh
scripts/deploy-uno-q-wifi.sh arduino@UNO-Q-NAME.local
```

The script preserves private `data/`, keeps PowerGlove Vision as the default startup app, restarts the container, and verifies the Learn, Debug, and secure Setup pages. If mDNS is temporarily unavailable, use the UNO Q's current IP address. Matrix firmware updates remain a separate App Lab operation; USB is the safest recovery route.

Install the safe-shutdown helper (required)

The helper requests `systemctl --no-block halt`, not `poweroff`: the UNO Q can reboot after a power-off request. A graceful halt stops Linux without requesting another boot; it does not disconnect electrical power. LEDs may remain lit. **Known limitation, confirmed September 3, 2026:** our UNO Q restarted after reaching the halt target both through a powered USB-C hub and when connected directly to a Mac. Shutdown is therefore not a verified way to keep this board stopped. Do not treat the website disappearing as a safe-to-unplug indicator or rely on a fixed countdown. See [Arduino shutdown guidance](#).

The App Lab container is intentionally unprivileged and cannot halt the Linux host. Complete the standard installation by installing the supplied fixed-purpose systemd path helper once:

```sh
scripts/install-uno-q-shutdown-helper.sh arduino@UNO-Q-NAME.local
```

Enter the UNO Q `arduino` account password at the remote `sudo` prompt. The script does not read or store it. The helper watches only the fixed `data/shutdown-request` path and can perform only a system halt. After it is installed, **Shutdown** is available on Dashboard and Setup. Each press requires browser confirmation and warns that power must be restored or cycled to restart the UNO Q. Its boot-time tmpfiles rule recreates the readiness marker if the UNO Q reboots or App Lab replaces the application directory.

Verify the helper without triggering shutdown:

```sh
ssh arduino@UNO-Q-NAME.local
systemctl is-enabled powerglove-system-shutdown.path
systemctl is-active powerglove-system-shutdown.path
exit
```

Expected results are `enabled` and `active`. On the validated cabinet these results, the readiness marker, both live buttons, and rejection of unconfirmed requests were verified on September 3, 2026. Do not test the accepted API path unless you intend to shut down the UNO Q.

RetroPie updates

```sh
cd /opt/powerglove-src
sudo git pull --ff-only
sudo /opt/powerglove/bin/python -m pip install -e '/opt/powerglove-src[receiver]'
sudo systemctl restart powerglove-receiver.service
```

Back up a customized `/etc/powerglove/games.json` before replacing it.

Duplicate App Lab entries

Importing a newer ZIP may create a timestamped app instead of replacing the old one. Verify the new app and its private settings first. Enable **Run at startup** for only one copy, stop the older copy, then remove it through App Lab.

Troubleshooting

Matrix shows a blinking X

- Confirm that an active gesture profile is selected. **Gestures off** should display the animated glove attract sequence, never the error X.
- Confirm the camera is connected through the powered hub.
- Try another hub port or USB cable.
- Check whether Linux sees a USB camera; internal `qcom-venus-encoder` and `qcom-venus-decoder` nodes are codecs, not your camera.
- Restart the app after checking power and cabling.

Camera appears only after reconnecting it

This usually indicates USB enumeration or power trouble. Keep the powered hub energized before starting the UNO Q, try another cable, and avoid passive adapters. The app itself waits for a camera and should recover when it appears.

First start takes several minutes

Expected: the UNO Q downloads its private Python 3.12 runtime and vision libraries once. Later launches reuse the persistent cache. Keep internet access available and watch the App Lab log for progress.

Setup page does not open

- Ordinary settings: `http://UNO-Q-NAME.local:8088/setup`
- Secure pairing: `https://UNO-Q-NAME.local:8443/setup`
- Try the board's IP address if `.local` does not resolve.
- HTTPS and HTTP are not interchangeable on these ports.

Password pairing fails

- Prepare a new attempt and use its new matrix PIN.
- Confirm the RetroPie username and password can log in through SSH.
- The account must be allowed to run `sudo` with that password.
- Prefer the one-time-code method if password SSH is disabled.

Controller does not appear on RetroPie

```sh
SH
systemctl is-enabled powerglove-receiver.service
systemctl is-enabled powerglove-receiver.timer
sudo systemctl status powerglove-receiver.service
sudo systemctl status powerglove-receiver.timer
sudo journalctl -u powerglove-receiver.service -n 100 --no-pager
ls -l /dev/uinput
```

Allow 45 seconds after boot. Confirm `uinput` is loaded and `/etc/powerglove/token` is not empty. Expected boot enablement is `disabled` for the service and `enabled` for the timer. The virtual controller appears only after an authenticated packet arrives.

Frontend slowdown or USB-device conflicts

Confirm the receiver service was not enabled directly. Stop it, restart EmulationStation, and start the receiver afterward as an A/B test. If the frontend becomes responsive, restore the supplied timer. Also ensure Pixelcade's `game-select` and `system-select` directories contain only one executable hook each if you use a BitPixel display; executable backup scripts are additional hooks.

Controller exists but does not move

- Select **Start controller** on Setup or Debug.
- Confirm a calibrated hand and controller output on Debug.
- Verify the receiver address and UDP 55355 connectivity.

- Check whether an existing cabinet input merger filters the virtual device.

Profiles do not change

- Test `powerglove-profile` manually.
- Check `uno_q` and `token_file` in `/etc/powerglove/launcher.json`.
- Confirm both runcommand hooks call the supplied helper scripts.
- Match the exact ROM basename in `/etc/powerglove/games.json`.

FAQ: What if the console name cannot be resolved?

1. In **Connection**, enter your console's actual hostname, such as `RETROPIE-NAME.local`, then select **Test console name**. Use a hostname or IPv4 address, not `http://`, a port, or a page path. This tests resolution from the UNO Q app; successful lookup on your laptop alone is not sufficient.
2. Confirm the RetroPie console is powered on and connected to your LAN. On its terminal, run `hostname` and `hostname -I` to confirm its name and current addresses. Do not assume an old DHCP address is still correct.
3. From the PowerGlove source directory on RetroPie, run `sudo python3 scripts/setup-machine.py retropie --check`. Check Avahi with `systemctl is-active avahi-daemon` and `systemctl is-enabled avahi-daemon`. If setup is incomplete, rerun `sudo python3 scripts/setup-machine.py retropie --peer UNO-Q-NAME.local`, using your board's actual name, and review every FAIL or ACTION result.
4. On the UNO Q, from the app directory, run `sudo python3 scripts/setup-machine.py uno-q --check`. This checks the configured destination from inside the application. If installation is incomplete, rerun `sudo python3 scripts/setup-machine.py uno-q`. Do not manually patch `.cache/app-compose.yaml`: App Lab regenerates it. The shipped resolver brick supplies the persistent configuration.
5. Check that both machines are on a network that allows communication between devices. Guest Wi-Fi, client isolation, VPN routing, separate VLANs, or multicast filtering can prevent `.local` discovery. mDNS uses UDP port 5353; do not disable your firewall wholesale or expose the app to the Internet to fix discovery.
6. As a diagnostic or fallback, enter RetroPie's current LAN IPv4 address in **Connection** and test again. If that works while the name fails, investigate mDNS. For continued use, reserve that address in your router so DHCP does not change it. Save the intended destination using the normal Connection workflow; changing the address does not replace pairing credentials. If RetroPie also contacts the UNO Q by name, check that reverse direction separately.
7. If neither name nor IP works, investigate connectivity and the service itself, not just Avahi. A successful name test only establishes name resolution; pairing, the receiver, controller output, and emulator mappings must also work. Retry after boot has finished, then collect the exact error and setup-check results if it still fails. Never share tokens, passwords, or private SSH keys.

After fixing the problem, reboot both machines and repeat **Test console name** before testing gameplay. The app-owned resolver has been verified across a UNO Q reboot and a changed RetroPie DHCP address; no fixed IP entry is required for `.local` use.

Uninstalling

On RetroPie:

```sh
SH
sudo systemctl disable --now powerglove-receiver.service
sudo rm /etc/systemd/system/powerglove-receiver.service
sudo systemctl daemon-reload
```

Remove only the PowerGlove lines from the runcommand hooks. After backing up custom profiles, `/opt/powerglove`, `/opt/powerglove-src`, and `/etc/powerglove` may be removed manually.

On the UNO Q, stop the app, disable **Run at startup**, and remove it through Arduino App Lab. Its private `data` directory contains the device token and cached runtime.

Remove the host shutdown helper separately:

```sh
sudo systemctl disable --now powerglove-system-shutdown.path
sudo rm /etc/systemd/system/powerglove-system-shutdown.path \
  /etc/systemd/system/powerglove-system-shutdown.service \
  /etc/tmpfiles.d/powerglove-system-shutdown.conf
rm -f /home/arduino/ArduinoApps/powerglove-vision/data/.shutdown-enabled
sudo systemctl daemon-reload
```

Project note

PowerGlove Vision is an independent MIT-licensed hobbyist project by Iain Bennett. Nintendo, NES, Power Glove, Bad Street Brawler, Super Glove Ball, and other marks belong to their respective owners. No ROM images are distributed with this project. Third-party runtime components retain their own licenses and terms as documented in THIRD_PARTY_COMPONENTS.md.

Local hostnames survive App Lab recreation

The `local:avahi_resolver` brick in `app.yaml` starts a small resolver service whenever App Lab starts the app. It mounts the host Avahi directory read-only and accepts only bounded IPv4 `.local` lookups over `data/.avahi-resolver.sock`. No network port, Docker socket, host networking, or privileged mode is used. The main app uses this private socket for gameplay and pairing. Answers expire after five seconds; IP addresses are not pinned.

Run the one-command UNO Q setup to enable host Avahi, then start the app through App Lab. Connection's hostname test should resolve `RETROPIE-NAME.local`. The service is part of the app, not a manual edit to generated `.cache` files. App Lab regeneration was tested successfully; a complete board reboot remains the final hardware check. A temporary missing Avahi socket returns a retryable error and does not stop the resolver service.